高数B上. HW3.

习题2.5.1.(2). $(\sqrt{x+2}-\sqrt{2})\sin x = \frac{x}{\sqrt{x+2}+\sqrt{2}}\sin x$

$\lim_{x\to 0} \frac{x\sin x}{\sqrt{x+2}+\sqrt{2}} \cdot \frac{1}{x^2} = \frac{1}{2\sqrt{2}}$ 故为 $=$ 阶小量.

2. 证明: $\lim_{x\to 0} \frac{\alpha(x)}{x} = \lim_{x\to 0} x \cdot \lim_{x\to 0} \frac{\alpha(x)}{x^2} = 0\cdot 0 = 0.$

故 $\alpha(x) = o(x)$. □.

4.(2) $d(1+x)^{\alpha}\Big|_{x=0} = \alpha(1+x)^{\alpha-1}dx\Big|_{x=0} = \alpha dx.$ $(\alpha>0)$.

6. $f(x) = \frac{2}{x-1}$. $f(3) = 1$. $f(3.001) = \frac{2}{2.001}$

$f'(x) = -\frac{2}{(x-1)^2}$. $f'(3) = -\frac{1}{2}$.

故 $\Delta y = \frac{2}{2.001} - 1 = -\frac{1}{2001}$. 微分是 $dy = -\frac{1}{2}dx = -\frac{1}{2000}$.

7. 令 $f(x) = \sqrt[5]{x}$. $f(2) = 32$.

$f'(x) = \frac{1}{5}x^{-\frac{4}{5}}$. $f'(32) = \frac{1}{5}\times\frac{1}{16} = \frac{1}{80}$.

则 $f(32.16) \approx f(32) + 0.16 f'(32) = 2 + \frac{0.16}{80} = 2.002$

8.(3) 左边求导为 $\frac{1}{1+(\frac{y}{x})^2}\cdot\frac{y'x-y}{x^2} = \frac{xy'-y}{x^2+y^2}$

右边求导为 $\frac{1}{2}\cdot\frac{1}{x^2+y^2}\cdot(2x+2yy') = \frac{x+yy'}{x^2+y^2}$

因此, $xy'-y = x+yy' \Rightarrow y' = \frac{x+y}{x-y}$.

9.(2) $(e^{xy})' = e^{xy}\cdot(xy)' = e^{xy}(y+xy')$

$(5x^2y)' = 5(x^2)'y + 5x^2y' = 10xy + 5x^2y'$

$\Rightarrow e^{xy}(y+xy') = 10xy + 5x^2y'$

代入 $x = \frac{e^2}{10}$, $y = \frac{20}{e^2} \Rightarrow e^2(\frac{20}{e^2} + \frac{e^2}{10}y') = 20 + \frac{e^4}{20}y'$

即 $y' = 0$.

No.

Date

10.(3) $\frac{dx}{dt} = -\frac{1}{\sqrt{1-\frac{1}{1+t^2}}} \cdot (-\frac{1}{2})(1+t^2)^{-\frac{3}{2}} \cdot 2t = \frac{t}{|t|}\frac{1}{1+t^2} = \frac{\operatorname{sgn} t}{1+t^2}$

$\frac{dy}{dt} = \frac{1}{\sqrt{1-\frac{t^2}{1+t^2}}} \cdot \cdot \frac{\sqrt{1+t^2}-\frac{t^2}{\sqrt{1+t^2}}}{1+t^2} = \frac{1}{1+t^2}$.

$\Rightarrow y' = \frac{dy/dt}{dx/dt} = \frac{1}{\operatorname{sgn} t} = \operatorname{sgn} t$.

习题2.6.1.(4) $y = \frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1}$.

由于 $(\frac{1}{x})^{(n)} = \frac{(-1)^n \cdot n!}{x^{n+1}}$, $(\frac{1}{x+1})^{(n)} = \frac{(-1)^n \cdot n!}{(x+1)^{n+1}}$.

故 $y^{(n)} = (-1)^n \cdot n! \left(\frac{1}{x^{n+1}} - \frac{1}{(x+1)^{n+1}}\right)$.

7. $f^{(k)}(x) = \frac{n(n+1)\cdots(n+k-1)}{(1-x)^{n+k}}$, 故 $f^{(k)}(0) = n(n+1)\cdots(n+k-1)$.

8. $y = x^2 \ln(1+x)$

$y^{(50)} = \sum_{k=0}^{50} C_{50}^k (x^2)^{(k)} (\ln(1+x))^{(50-k)}$

$= x^2 (\ln(1+x))^{(50)} + 50 \cdot 2x \cdot (\ln(1+x))^{(49)} + 2450 \cdot 2 \cdot (\ln(1+x))^{(48)}$

$= x^2 \cdot \left(-\frac{49!}{(1+x)^{50}}\right) + 100x \cdot \frac{48!}{(1+x)^{49}} + 4950 \cdot \frac{47!}{(1+x)^{48}}$

习题2.7.2.

$\int (1+\sqrt{x})^2 dx = \int dx + 2\int \sqrt{x}\, dx + \int x\, dx$

$= 1 + \frac{4}{3}x^{3/2} + \frac{1}{2}x^2 + C$.

5. $\int \cot^2\varphi\, d\varphi = \int \frac{1-\sin^2\varphi}{\sin^2\varphi} d\varphi = \int \frac{d\varphi}{\sin^2\varphi} - \int d\varphi = -\cot\varphi - \varphi + C$.

6. $\int \frac{x^2+3}{1+x^2} dx = \int dx + 2\int \frac{dx}{1+x^2} = x + 2\arctan x + C$.

No.

Date

7. $\int \frac{3}{\sqrt{x}} + \frac{4}{\sqrt{1-x^2}}\,dx = 6\sqrt{x} + 4\arcsin x + C.$

9. $\int \frac{1-x}{1-\sqrt[3]{x}}\,dx = \int 1+\sqrt[3]{x}+x^{\frac{2}{3}}\,dx = x + \frac{3}{4}x^{\frac{4}{3}} + \frac{3}{5}x^{\frac{5}{3}} + C.$

12. $\int (2\cosh x - \sinh x)\,dx = \int (e^x + e^{-x} - \frac{e^x - e^{-x}}{2})\,dx$

$= \int \frac{1}{2}e^x + \frac{3}{2}e^{-x}\,dx = \frac{1}{2}e^x - \frac{3}{2}e^{-x} + C.$

14. $\int \frac{dx}{\sin^2 x \cos^2 x} = \int \frac{1}{\sin^2 x} + \frac{1}{\cos^2 x}\,dx = \tan x - \cot x + C.$

15. $\int \frac{2^{x+1}+3^{x-2}}{6^x}\,dx = 2\int (\frac{1}{3})^x dx + \frac{1}{9}\int (\frac{1}{2})^x dx$

$= -\frac{2}{\ln 3}\cdot 3^{-x} - \frac{1}{9\cdot \ln 2}\cdot 2^{-x} + C.$

16. $\int \frac{dx}{x^2(1+x^2)} = \int \frac{dx}{x^2} - \int \frac{dx}{1+x^2} = -\frac{1}{x} - \arctan x + C.$